

## Section1\_2.R

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```
### Section 1.2 Examples ###
```

```
temp<-c(87,80,91,60,91,77,87,77,80,78,89)
length(temp)
```

```
## [1] 11
```

```
mean(temp)
```

```
## [1] 81.54545
```

```
var(temp)
```

```
## [1] 81.67273
```

```
sd(temp)
```

```
## [1] 9.037296
```

```
# Convert from F to C
```

```
temp_c<-0.556*temp-17.778
length(temp_c)
```

```
## [1] 11
```

```
mean(temp_c)
```

```
## [1] 27.56127
```

```
var(temp_c)
```

```
## [1] 25.24798
```

```
sd(temp_c)
```

```
## [1] 5.024737
```

```
temp_sort<-sort(temp)
temp_sort
```

```
## [1] 60 77 77 78 80 80 87 87 89 91 91
```

```
a<-c(60,77,77,78,80,80,87,87,89,91,91)
```

```
temp_sort==a
```

```
## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

```
median(a)
```

```
## [1] 80
quantile(a,type=2)

##    0%   25%   50%   75%  100%
##    60    77    80    89    91
IQR(a,type=2)

## [1] 12
# What is type, use R help to find out.
?quantile

## starting httpd help server ... done
# Default type for quantile() and IQR() is type=7.
quantile(a)

##    0%   25%   50%   75%  100%
## 60.0 77.5 80.0 88.0 91.0
IQR(a)

## [1] 10.5
b<-c(36,38,39,39,40,42,45,45,48,48,52,62)
median(b)

## [1] 43.5
quantile(b,type=2)

##    0%   25%   50%   75%  100%
## 36.0 39.0 43.5 48.0 62.0
IQR(b,type=2)

## [1] 9
# Feel free to use the default type=7.
quantile(b)

##    0%   25%   50%   75%  100%
## 36.0 39.0 43.5 48.0 62.0
IQR(b)

## [1] 9
# The following shows how to import a data file from external source.

ClassMajor <- read.csv("C:/Users/jiewa/Downloads/ClassMajor.csv")
View(ClassMajor)

# Using Base R.

table(ClassMajor$Class)

##
## Freshman      Junior      Senior Sophomore
```

```
##           4           51           28           36
table(ClassMajor$Major)

##
##   Applied and Computational Mathematics           Applied Physics
##                                     6                               1
##                               Biochemistry           Chemical Engineering
##                                     13                             26
##                               Chemistry           Civil Engineering
##                                     2                               8
##                               Computer Engineering           Computer Science
##                                     4                               2
##                               Cybersecurity           Electrical Engineering
##                                     7                             17
##                               Machine Learning           Manufacturing Engineering
##                                     1                               9
##                               Mathematics           Mathematics Education Premajor
##                                     8                               1
##                               Mechanical Engineering Mechanical Engineering Pre-Professional
##                                     3                               3
##                               Middle East Studies / Arabic           Open-Major
##                                     1                               1
##                               Physics           Physics and Astronomy
##                                     3                               1
##                               Software Engineering
##                                     2

# Using tidyverse to summarize the data.
# Installing and using additional packages are optional. You are welcome to try
# these packages. In most cases, Base R should be sufficient.

library(tidyverse)

## Warning: package 'tidyverse' was built under R version 4.1.2

## -- Attaching packages ----- tidyverse 1.3.1 --
## v ggplot2 3.3.3    v purrr 0.3.4
## v tibble 3.1.2     v dplyr 1.0.7
## v tidyr 1.1.3      v stringr 1.4.0
## v readr 1.4.0      v forcats 0.5.1

## Warning: package 'tidyr' was built under R version 4.1.1
## Warning: package 'purrr' was built under R version 4.1.1
## Warning: package 'dplyr' was built under R version 4.1.1
## Warning: package 'forcats' was built under R version 4.1.1

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()

ClassMajor %>% count(Class)

##      Class  n
## 1 Freshman  4
## 2 Junior 51
```

```
## 3 Senior 28
## 4 Sophomore 36
```

```
ClassMajor %>% count(Major,sort=T)
```

```
## Major n
## 1 Chemical Engineering 26
## 2 Electrical Engineering 17
## 3 Biochemistry 13
## 4 Manufacturing Engineering 9
## 5 Civil Engineering 8
## 6 Mathematics 8
## 7 Cybersecurity 7
## 8 Applied and Computational Mathematics 6
## 9 Computer Engineering 4
## 10 Mechanical Engineering 3
## 11 Mechanical Engineering Pre-Professional 3
## 12 Physics 3
## 13 Chemistry 2
## 14 Computer Science 2
## 15 Software Engineering 2
## 16 Applied Physics 1
## 17 Machine Learning 1
## 18 Mathematics Education Premajor 1
## 19 Middle East Studies / Arabic 1
## 20 Open-Major 1
## 21 Physics and Astronomy 1
```

```
ClassMajor %>% count(Class,Major,sort=T)
```

```
## Class Major n
## 1 Sophomore Chemical Engineering 13
## 2 Junior Biochemistry 8
## 3 Junior Electrical Engineering 8
## 4 Junior Chemical Engineering 6
## 5 Junior Manufacturing Engineering 6
## 6 Senior Chemical Engineering 6
## 7 Junior Civil Engineering 5
## 8 Senior Electrical Engineering 5
## 9 Junior Cybersecurity 4
## 10 Sophomore Electrical Engineering 4
## 11 Junior Computer Engineering 3
## 12 Senior Mathematics 3
## 13 Sophomore Applied and Computational Mathematics 3
## 14 Sophomore Biochemistry 3
## 15 Sophomore Mathematics 3
## 16 Junior Mathematics 2
## 17 Junior Physics 2
## 18 Senior Applied and Computational Mathematics 2
## 19 Senior Biochemistry 2
## 20 Senior Cybersecurity 2
## 21 Senior Mechanical Engineering 2
## 22 Sophomore Computer Science 2
## 23 Sophomore Manufacturing Engineering 2
## 24 Sophomore Mechanical Engineering Pre-Professional 2
```

|       |           |                                         |   |
|-------|-----------|-----------------------------------------|---|
| ## 25 | Freshman  | Chemical Engineering                    | 1 |
| ## 26 | Freshman  | Chemistry                               | 1 |
| ## 27 | Freshman  | Civil Engineering                       | 1 |
| ## 28 | Freshman  | Open-Major                              | 1 |
| ## 29 | Junior    | Applied and Computational Mathematics   | 1 |
| ## 30 | Junior    | Chemistry                               | 1 |
| ## 31 | Junior    | Mathematics Education Premajor          | 1 |
| ## 32 | Junior    | Mechanical Engineering                  | 1 |
| ## 33 | Junior    | Mechanical Engineering Pre-Professional | 1 |
| ## 34 | Junior    | Middle East Studies / Arabic            | 1 |
| ## 35 | Junior    | Software Engineering                    | 1 |
| ## 36 | Senior    | Applied Physics                         | 1 |
| ## 37 | Senior    | Civil Engineering                       | 1 |
| ## 38 | Senior    | Manufacturing Engineering               | 1 |
| ## 39 | Senior    | Physics                                 | 1 |
| ## 40 | Senior    | Physics and Astronomy                   | 1 |
| ## 41 | Senior    | Software Engineering                    | 1 |
| ## 42 | Sophomore | Civil Engineering                       | 1 |
| ## 43 | Sophomore | Computer Engineering                    | 1 |
| ## 44 | Sophomore | Cybersecurity                           | 1 |
| ## 45 | Sophomore | Machine Learning                        | 1 |

#### # Homework Help

```

A<-c(18.0, 18.0, 18.0, 20.0, 22.0, 22.0, 22.5, 23.0, 24.0, 24.0, 25.0, 25.0, 25.0, 25.0, 26.0, 26.4)
B<-c(18.8, 18.9, 18.9, 19.6, 20.1, 20.4, 20.4, 20.4, 20.4, 20.5, 21.2, 22.0, 22.0, 22.0, 22.0, 23.6)
C<-c(20.2, 20.5, 20.5, 20.7, 20.8, 20.9, 21.0, 21.0, 21.0, 21.0, 21.0, 21.5, 21.5, 21.5, 21.5, 21.6)
D<-c(20.0, 20.0, 20.0, 20.0, 20.2, 20.5, 20.5, 20.7, 20.7, 20.7, 21.0, 21.1, 21.5, 21.6, 22.1, 22.3)

```